

Ex.No.6: 5G NR Frame Structure Analysis with Different Subcarrier Spacing

Objective 1:

```
clc;
```

```
clear;
```

```
close all;
```

```
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
```

```
mu = 0:4; % Numerology Index
```

```
bar(scs)
```

```
grid on
```

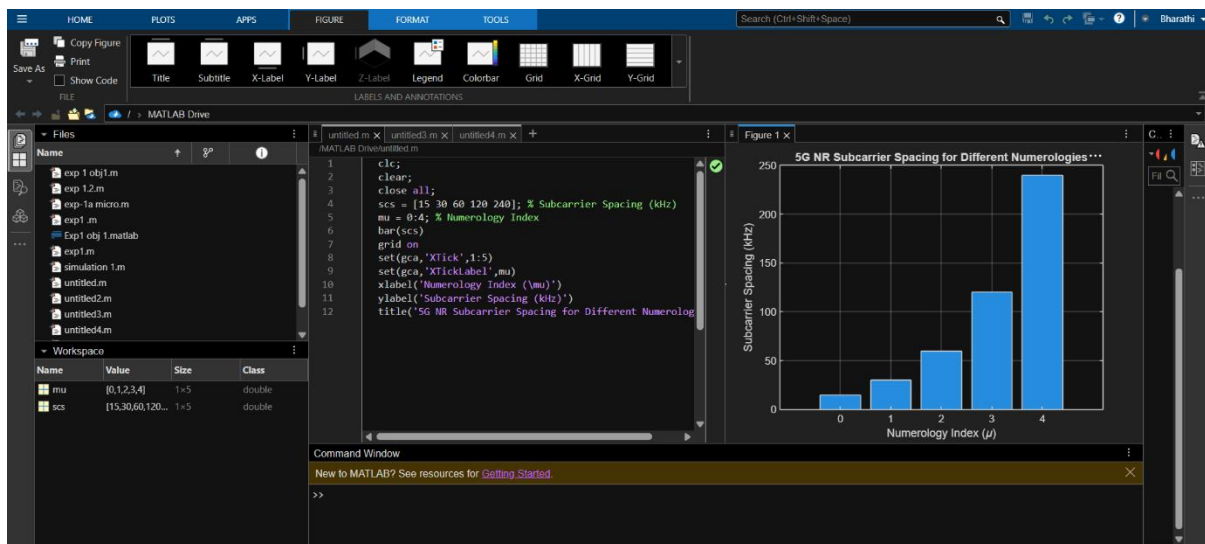
```
set(gca,'XTick',1:5)
```

```
set(gca,'XTickLabel',mu)
```

```
xlabel('Numerology Index (\mu)')
```

```
ylabel('Subcarrier Spacing (kHz)')
```

```
title('5G NR Subcarrier Spacing for Different Numerologies')
```



Objective 2:

```
clear;
```

```
close all;
```

```
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz) slot = [1 0.5 0.25  
0.125 0.0625]; % Slot Duration (ms)
```

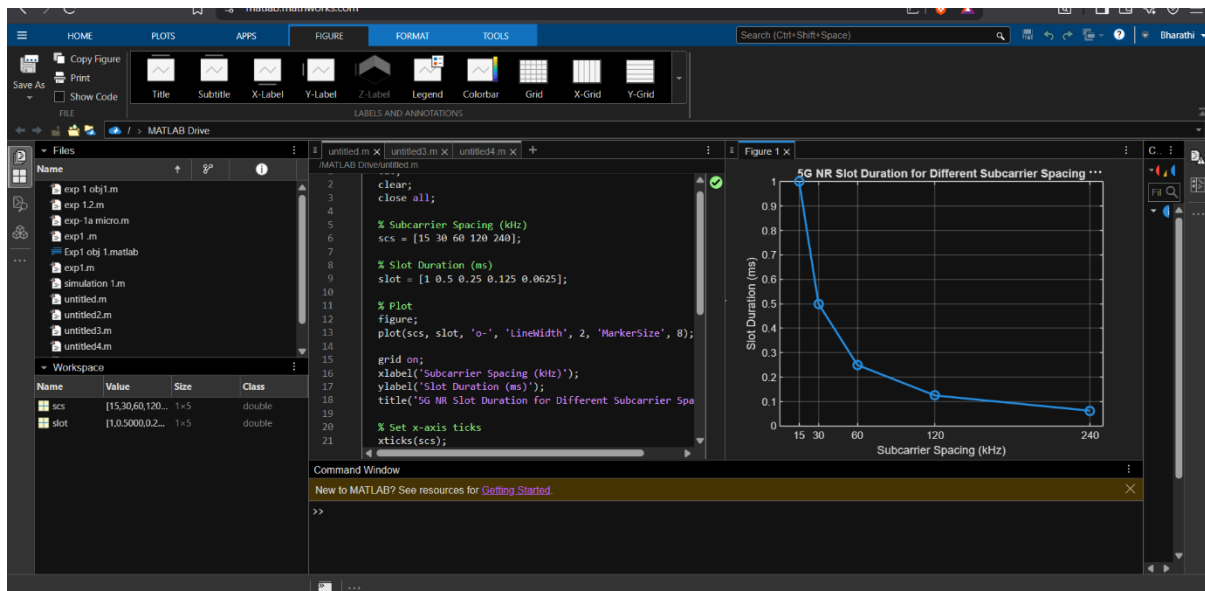
```
plot(scs, slot, 'o-', 'LineWidth', 2)
```

```
grid on
```

```
xlabel('Subcarrier Spacing (kHz)')
```

```
ylabel('Slot Duration (ms)')
```

```
title('5G NR Slot Duration for Different Subcarrier Spacing')
```



Objective 3:

```
clc;
```

```
clear;
```

```
close all;
```

```
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
```

```
symbols = [14 14 14 14 14]; % OFDM Symbols per Slot
```

```
figure
```

bar(scs, symbols)

grid on

xlabel('Subcarrier Spacing (kHz)')

ylabel('Number of OFDM Symbols')

title('OFDM Symbols per Slot for Different Subcarrier Spacing')

ylim([0 16])

